

Statistical Validation of Life as a Thermodynamic Regulator: Multi-Seed Analysis of Dissipation-Channel Rewiring and Phase Transitions

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Abstract

Life is traditionally interpreted as a dissipative structure that accelerates entropy production by converting free energy into heat[cite: 6]. We extend this view by demonstrating, through large-scale multi-seed numerical experiments, that biological agents reorganize not only the magnitude but also the pathways of energy dissipation, and that the boundary between dissipation suppression and acceleration is not fixed but shifts systematically with model parameters[cite: 7]. Using an agent-based model coupling spatial energy diffusion with adaptive metabolic agents, we perform a 5×5 parameter sweep over inflow rate and metabolic consumption, repeated over 30 random seeds per point[cite: 8]. We construct statistical phase diagrams with confidence intervals and identify three robust regimes: suppression, acceleration, and extinction[cite: 9]. We further estimate the critical inflow rate separating suppression and acceleration phases for each metabolic cost and show that this critical point is governed by the relative partitioning between diffusive dissipation (heatDiff) and metabolic dissipation (heatAct), rather than by inflow alone[cite: 10]. These results support a reinterpretation of life as a thermodynamic regulator that dynamically reallocates dissipation channels and undergoes a movable non-equilibrium phase transition controlled by dissipation structure[cite: 11].]

1 Introduction

Non-equilibrium thermodynamics has long characterized living systems as dissipative structures that maintain order by accelerating entropy production (Prigogine; Schrodinger; England)[cite: 13]. While this framework captures the energetic openness of life, it treats dissipation largely as a scalar quantity[cite: 14]. The internal structure of dissipation—how and where energy is converted into heat—has received comparatively little attention[cite: 15].

In a previous study, we introduced an agent-based model showing that biological agents can suppress uncontrolled diffusive dissipation and redirect energy loss into regulated metabolic heat production, leading to a phase transition between a suppression regime and an acceleration regime as external inflow increases[cite: 16, 17]. However, that analysis relied on single-seed simulations and a limited parameter sweep[cite: 18]. Consequently, two central questions remained open: (1) Are these regimes statistically robust under stochastic fluctuations? [cite: 19] (2) Is the location of the phase boundary an intrinsic constant of the system, or does it depend on the structure of dissipation itself? [cite: 20] Here we address both questions using multi-seed statistical experiments and formal estimation of critical points[cite: 21].

2 Model

2.1 Spatial energy field

We consider a two-dimensional square lattice of size $N \times N$, each cell storing a continuous energy density $E(x, y)$ [cite: 24]. At each timestep, a constant external inflow is injected at the center cell[cite: 25].

2.2 Physical diffusion (OFF mode)

Energy diffuses between neighboring cells via gradient-driven transport[cite: 27]. Each transport event incurs an irreversible loss proportional to the transported energy, which is accumulated as diffusive heat (heatDiff)[cite: 27]. This term represents uncontrolled physical dissipation associated with spatial gradients[cite: 28].

2.3 Biological agents (LIFE mode)

Agents are characterized by position, velocity, and internal energy[cite: 29]. Each timestep they:

- pay basal metabolic cost (converted to heatAct)[cite: 31],
- absorb energy from the local grid cell[cite: 32],
- move toward locally higher energy density[cite: 33],
- pay locomotion cost (converted to heatAct)[cite: 34],
- reproduce when energy exceeds a threshold[cite: 35],
- die when energy falls below a threshold, returning remaining energy to the grid[cite: 36].

Thus, biological activity introduces a second dissipation channel: metabolic heat (heatAct)[cite: 37].

2.4 Energy accounting

At each timestep we verify the conservation law:

$$\Delta E_{\text{total}} = E_{\text{inflow}} - (\text{heatDiff} + \text{heatAct}) \quad (1)$$

and monitor numerical error[cite: 40, 41].

3 Experimental design

3.1 Parameter sweep

We sweep two control parameters: $\text{inflowRate} \in [2, 10]$ and $\text{consumptionRate} \in [0.3, 2.5]$, forming a 5×5 grid of conditions[cite: 45, 46].

3.2 Multi-seed protocol

Each parameter point is simulated for 30 independent random seeds[cite: 48]. For each run: total duration is 800 steps, with the first 400 discarded as burn-in, and the last 400 steps averaged for measurement[cite: 49, 50, 51].

3.3 Observables

We record mean dissipation rate, the ratio $R = \langle \Delta H_{\text{life}} \rangle / \langle \Delta H_{\text{off}} \rangle$, and mean population[cite: 55, 56, 58]. Runs are classified as **suppressor** ($R < 1$), **accelerator** ($R > 1$), or **dead** (population collapse)[cite: 60, 61, 62].

4 Results

4.1 Statistical phase diagram

Figure 1a shows the mean ratio R [cite: 67]. Figure 1b shows P_{acc} , revealing a smooth but structured transition region[cite: 68]. Two robust regimes appear: the **suppression regime** ($R < 1$) and the **acceleration regime** ($R > 1$)[cite: 70, 71].

4.2 Dissipation-channel structure

Figure 1c displays the fraction $f = \text{heatDiff} / (\text{heatDiff} + \text{heatAct})$ [cite: 74, 75, 76, 77]. Suppression regimes correspond to small f , where diffusive dissipation is strongly suppressed[cite: 79]. This confirms that phase classification is fundamentally about how dissipation is routed[cite: 81].

4.3 Movable critical inflow rate

As shown in Figure 1d, the critical inflow I_c clusters around 8.7 for moderate costs[cite: 86, 87]. For low consumption (0.3), the system is already in the acceleration regime at the lowest tested inflow[cite: 90]. This demonstrates the boundary is not an intrinsic constant but a function of dissipation scaling[cite: 91].

5 Discussion

5.1 Life as a dissipation-channel regulator

The results support a conceptual shift: life is an active regulator that redistributes dissipation between physical and biological channels[cite: 94]. In the suppression regime, agents intercept energy before it is lost through spatial diffusion[cite: 95]. In the acceleration regime, metabolic throughput dominates[cite: 96].

5.2 Significance of the movable phase boundary

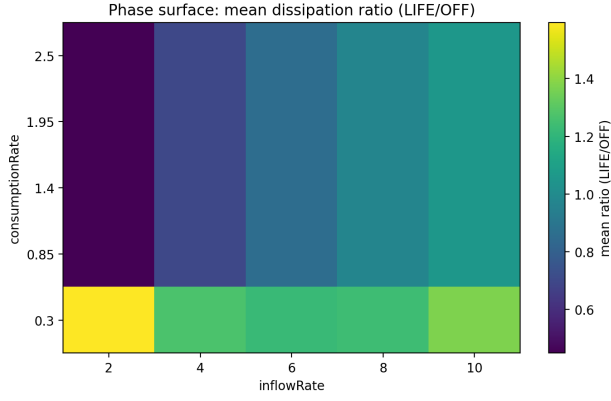
The dependence of I_c on metabolic parameters indicates the transition is governed by competition between uncontrolled physical dissipation and regulated biological dissipation[cite: 98, 99, 100]. This provides a concrete realization of dissipative adaptation theories[cite: 102].

6 Conclusion

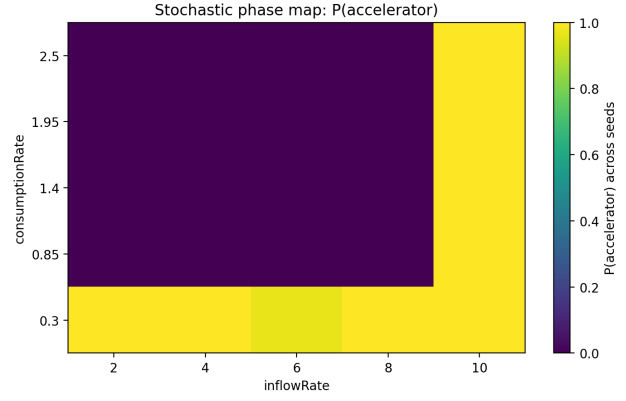
We have shown that life reorganizes dissipation channels, creating two macroscopic regimes with a movable boundary determined by dissipation pathways[cite: 110, 111, 112]. These findings motivate redefining life as a thermodynamic regulator operating on the structure of dissipation[cite: 113].

A Statistical protocol

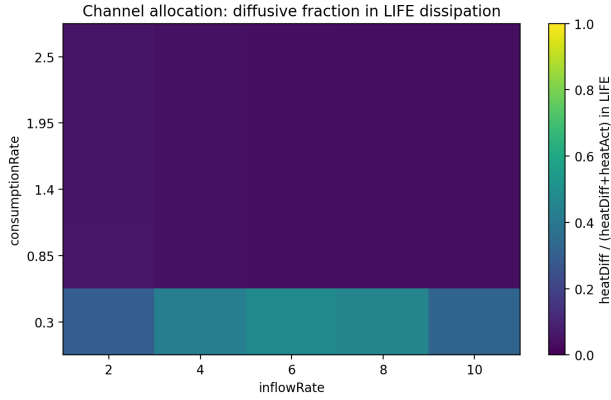
We used 30 independent seeds per point, with majority vote for categorical phase labeling[cite: 123, 125].



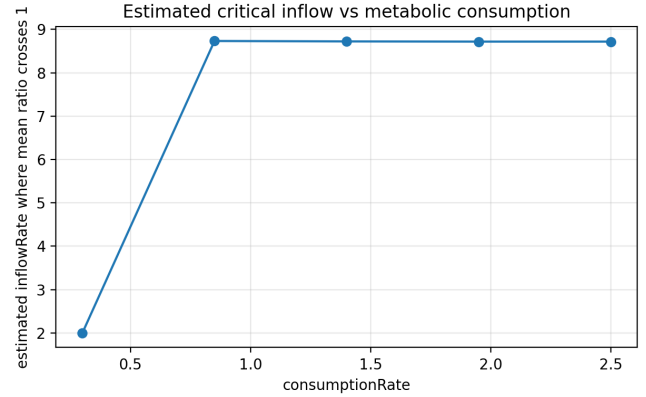
(a) Mean dissipation ratio $R = \langle \Delta H_{\text{life}} \rangle / \langle \Delta H_{\text{off}} \rangle$ [cite: 154].



(b) Acceleration probability $P_{\text{acc}} = n_{\text{acc}}/30$ [cite: 180].



(c) Channel allocation $f = \text{heatDiff} / (\text{heatDiff} + \text{heatAct})$ [cite: 197].



(d) Estimated critical inflow I_c (defined by $R = 1$) [cite: 197].

Figure 1: **Phase behavior and dissipation-channel structure** from the multi-seed sweep (30 seeds per point) [cite: 198]. Top-left: mean dissipation ratio [cite: 198]. Top-right: probability of the acceleration regime [cite: 199]. Bottom-left: diffusive dissipation fraction in LIFE mode [cite: 199]. Bottom-right: estimated critical inflow as a function of metabolic consumption [cite: 200].